

Roll No.

TPE-601

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)**

END SEMESTER EXAMINATION, June, 2023

RESERVOIR MODELING AND SIMULATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain Calstic and Carbonate rocks in details, their composition and relative abundance on the earth form hydrocarbon point of view. (CO2)
- (b) Name the various types of physical properties of rock ? What are the three essential properties of the rocks to be a reservoir ? Explain in detail. (CO1)
- (c) What are the various types of model required for Exploration and Development of oil and gas field ? Name the industry leading software use for building these models. (CO4)

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2. (a) Explain Structural and stratigraphic traps in detail. Draw a cross section of a petroleum system and explain its all important components. (CO3)
- (b) Explain Reservoir Modeling, Reservoir Engineering and various step of Reservoir Engineering for better field development. (CO4)
- (c) Explain the term sequential workflow in context with the modeling process. What are the essential data required for building a reservoir model ? (CO5)
3. (a) Illustrate various drive mechanism of petroleum system with figure in details. How these drive mechanism help in primary recovery methods ? (CO2)
- (b) How packing and sorting effect the porosity and permeability ? Define with figure Cubical packing, orthorhombic-packing and rhombohedra-packing of spheres along with their porosity values. (CO4)
- (c) Design a suitable workflow for making a structure model and explain its various steps in detail. (CO3)
4. (a) Define facies. What is the difference between facies log and normal log ? Write the facies modeling workflow and define its each step in detail. (CO1)
- (b) Explain deterministic and Stochastic methods for Petrophysical modeling. Where these methods will be recommended ? (CO3)
- (c) What are the different methods for up-scaling the Petrophysical property Porosity, Saturation and Permeability. Give the example of each. (CO5)

(3)

5. (a) Explain the term variance, sill, nugget and range in variogram draw with diagram. how many type of variogram is used ? what are the uses of variogram ? (CO3)
- (b) Define the term reserve estimation. Write its uses and the formula for volumetric reserve estimation for an oil and gas field. (CO4)
- (c) What are the various techniques of reserve estimation ? Why it is important to estimate reserve at different stages ? (CO5)

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B. TECH. (PE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

DATA SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the role of data in analytics and the process of deriving insights from data. Discuss the importance of data-driven decision making in various industries and how analytics and data science contribute to organizational success ? (CO3)
- (b) Discuss the various stages :
 - (i) Data collection and pre-processing
 - (ii) Exploratory data analysis
 - (iii) Feature Engineering

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(iv) Model selection and training

(v) Evaluation

Explain how these frameworks help analysts tackle complex problems and make informed decisions based on data-driven insights. (CO3)

(c) Define the importance of model evaluation and selection in data science, and highlight considerations such as overfitting, bias-variance tradeoff, and model interpretability. (CO3)

2. (a) What are aggregate functions in SQL and its role in data analytics? Discuss common aggregate functions such as SUM, COUNT, AVG, MAX, and MIN. Explain how these functions facilitate calculations and summarization of data, enabling deeper insights and analysis. (CO2)

(b) Create a table named Orders with columns for order ID, customer ID, and total amount, this table should have at least 10 (ten) entries. Write a SQL query to retrieve the customer IDs of customers who have placed orders with a total amount greater than the average total amount of all orders. Include only customers who have placed at least two orders.

(CO2)

(c) Explain how ORDER BY clause allows data analysts to sort retrieved data in ascending or descending order based on one or more columns. Why is it necessary to sort retrieved data? (CO2)

3. (a) Using a specific dataset, demonstrate how you would create a combination chart in Excel, incorporating both a line chart and a bar

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B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023
NATURAL GAS ENGINEERING
Time : Three Hours
Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) Calculate the density and gas specific gravity of natural gas in the reservoir having the following composition :

Composition	Mole fractions
C1	0.785
C2	0.085
C3	0.023
CO ₂	0.030
H ₂ S	0.020

The reservoir is having at initial condition of $P_i = 3000$ psia and 180°F .
By calculating Sp. Gr. Of the gas also calculate T_{pc} , P_{pc} , T_{pr} , P_{pr} by

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using Standing and Katz correlations (take compressibility factor $Z = 0.87$ for the data given). (CO1, CO2)

- (b) What are the processes/methods of conventional natural gas production ? Write down the steps in well completion.

Solve the numerical :

A dry gas well is producing a gas stream of the following molar composition: 95 percent methane and 5 percent carbon dioxide. The molar mass of methane is 16g/mol and that of carbon dioxide is 44g/mol. Assuming Ideal gas behavior, gas constant $R = 8.31 \text{ J/mol K}$. Find the gas stream density at 10^7 Pa and 350 K in SI unit. (CO1, CO2)

- (c) What do you mean by clathrates ? Describe in detail about all the properties of gas hydrate.

Draw boundaries with respect to depth and temperature for different phase by stating the different region as methane gas, methane gas hydrates water and ice. (CO3)

2. (a) A gas well is producing a natural gas with the following composition:

Component	Y_i
CO_2	0.05
C_1	0.90
C_2	0.03
C_3	0.02

Assuming Ideal gas behavior, Calculate the following :

- Apparent molecular weight of the gas
- Gas specific gravity
- Gas density at 2000psi and 150 degree F
- Specific volume at 2000psi and 150 degree F
- Calculate Gas density at 2000psi and 150 degree F considering real gas by taking $Z = 0.85$ (CO3, CO4)

- (b) Explain hydraulic fracturing ? Discuss the hydraulic fracturing process, discuss in detail about the fracturing fluid by defining the base fluid, Energizing gases, gelling agent, cross linkers, surfactant and non-emulsifiers. Discuss the different types of proppant used during extraction of shale gas. (CO1, CO4)
- (c) What are the conventional natural gas production systems ? Write down the conditions in terms of pressure with respect to bubble point and dew point for the black oil, volatile oil, solution gas drive, wet gas and gas condensate with the help of phase diagram. (CO1, CO2)
3. (a) Classify and discuss the Sweetening process ? What are the different Amines used in the gas amine based sweetening process ? Define iron-sponge sweetening. Write down the reactions involved in the iron sponge process. (CO1)
- (b) Explain the following along with mathematical expression in the determination of water content of natural gas : (CO1, CO2)
- (i) The gravity correction factor
 - (ii) The salinity correction factor
 - (iii) Mole percent of H_2S equivalent concentration
 - (iv) Can water content be calculated by the dew point ?
- (c) Illustrate the dehydration systems. Discuss the phenomena absorption and adsorption process during the dehydration of natural gas. (CO2, CO5)

4. (a) Classify the Sweetening process. What are the different Amines used in the gas amine based sweetening process ? Define iron-sponge sweetening ? Write down the reactions involved in the iron sponge process. (CO1, CO5)
- (b) Why glycol/Amine process is used in sweetening of natural gas ? Discuss the proportions of solution being used in glycol/Amine process. What are the main advantages and disadvantages of glycol/Amine process in the sweetening of natural gas ? (CO4, CO5)
- (c) Draw the flow sheet diagram for the amine based sweetening process in the natural gas sweetening with their process descriptions mentioning the reactions occurred at absorber. (CO3, CO6)
5. (a) Draw the flow sheet diagram of Selexol process in the sweetening of natural gas with their process description. (CO6)
- (b) Draw the transportation system. Pipeline design equation in transportation of natural gas depends on which factor ? Write down the complex equations which have been developed for sizing natural gas pipelines in various flow conditions. (CO4, CO5)
- (c) Calculate the maximum quantity of gas that can be transported through 4" dia, 20 Km long horizontal pipeline under the following conditions :
- | | |
|----------------------------|-----------|
| Inlet pressure | 2000 psia |
| Maximum pressure drop | 500 psi |
| Average gas temp. | 100 F |
| Pipeline efficiency factor | 0.92 |
| Gas gravity | 0.6 |
| Zavg can be taken as | 0.9 |
- Use both Panhandle B and Weymouth methods. (CO2, CO3)

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**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)**

END SEMESTER EXAMINATION, June, 2023

PETROLEUM DRILLING ENGINEERING—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is difference between kick and kick-off point ? Discuss the differences and difficulties of deep vs. shallow kickoff points. (CO1)
- (b) Crookedness of wells is a serious problems in direction drilling, what are the cause of crookedness of holes ? What are the outcomes of crookedness of hole and possible remedies. Discuss. (CO4)
- (c) A dog-leg is expected at a depth of 5000 ft. A string of 5-in., 19.5 lb/ft, grade E drill pipe is being used. If the tensile load below this point is 200,000 lb, calculate the maximum dog-leg that can be allowed. (CO1)

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2. (a) What are the consideration in horizontal well profile design ?
Discuss. (CO2)
- (b) Calculate Torque while rotating off bottom in the horizontal well using following well data : (CO2)
- Build curve radius (R) = 1042 ft
Horizontal section = 1500 ft
WOB = 30,000 lbf
Mud weight = 9.2 ppg
BHA is made up mostly of HWDP
HWDP = 5", 50 lb/ft, OD of tool joint is 6.5"
- (c) Discuss the problems associated with drilling of horizontal wells.
3. (a) What is a multilateral well ? List the different types of multilateral well and draw their figure. What are the advantages of multilateral well ? What are the consideration in multilateral well planning ? (CO2)
- (b) What are cased hole and whipstock kick off in multilateral well ?
What are their advantages and disadvantages ? (CO2)
- (c) What are the flow control devices in multilateral well ? Discuss. (CO2)
4. (a) Discuss the collapse and burst criterion for casing design. (CO5)
- (b) During drilling of an 8.5 in hole at 8,000 ft, a complete loss of circulation was observed. Drilling was stopped and the mud level in the annulus was observed to fall rapidly. The well was filled with

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water of 8.3 ppg density until the annular level remained stationary. If the volume of water used was 65.7 bbl and mud density was 10 ppg, determine the formation pressure and the new mud weight required to balance the formation pressure. Assume the intermediate casing to be 9.625 in, 40# set at 6,000 ft. Drillpipe is Grade E, 5 in OD. (CO4)

- (c) What do you mean by lost circulation in drilling ? Describe causes of lost circulation and possible remedial methods. (CO4)
5. (a) The drill string becomes stuck while drilling a directional well. The driller reports that he can still circulate, but cannot rotate or reciprocate the pipe. (CO4)
- (i) What is the most likely cause of the stuck pipe ?
- (ii) What action should the driller take to free the pipe ?
- (b) Determine the size and number of drillcollars required to provide a weight-on-bit of 55,000 (CO6)
- lbf assuming
- Hole deviation = 0°
- Mud density = 12 pp
- (c) What are the criterion for drill pipe selection ? Discuss. (CO6)

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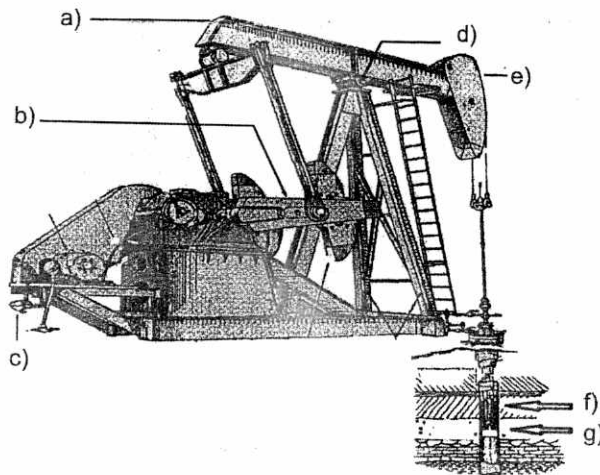
**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023
PETROLEUM PRODUCTION ENGINEERING—II**

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) Explain the load which are temporary in nature/on platform.(CO1, CO4)
(b) Describe the platform which rests on 4-6 pillar like legs. (CO1, CO4)
(c) From the given figure identify and explain type of Artificial lift and also identify parts (a, b, c, d, e, f, g) in the given figure. Also write why it is used as artificial lift. (CO1, CO4)



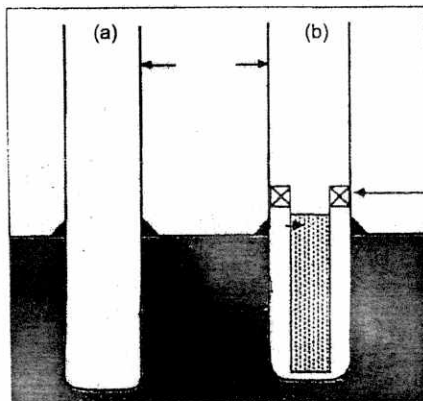
P. T. O.

2. (a) Explain the concept of offshore process equipment and open hole completion. (CO2, CO4, CO5)
- (b) Define EOR and also describe thermal method of EOR. (CO2, CO4, CO5)
- (c) Write 5-5 advantages and disadvantages of Gas lift, Rod Pumping and ESP. (CO2, CO4, CO5)
3. (a) Construct IPR of a vertical well in a saturated oil reservoir using Vogel's equation. The following data are given : (CO3)
- Porosity = 0.19
- Effective horizontal permeability = 8.2 md
- Pay zone thickness = 53 ft
- Reservoir pressure = 5651 psia
- Bubble-point pressure = 5561 psia
- Fluid formation volume factor = 1.1
- Fluid viscosity = 5 1.7 cp
- Total compressibility = 0.0000129 psi-1
- Drainage area = 640 acres ($r_e = 2980$ ft)
- Wellbore radius $r_w = 0.328$ ft
- Skin factor = 0
- (b) Suppose that 1000 bbl/day of 40°API, 1.2 cp oil is being produced through $2\frac{7}{8}$ in, 8.6-lb_m/ft tubing in a well that is 15° from vertical. If the tubing wall relative roughness is 0.001, calculate the pressure drop over 1000 ft of tubing. (CO3)

(3)

(c) Illustrate the nodes in an oil and gas production system and pressure trend along the flow path and also write all the abbreviations used for making nodes. Also describe what is nodal analysis and it is used in Petroleum Production. (CO3)

4. (a) From the given figure identify types of liner and differentiate between them and also identify and name the parts of liner which are shown by arrows. (CO2, CO5)



(b) Describe the concept of oil processing module and drill ships.

(CO2, CO5)

(c) Differentiate Between Gas injection and Water Injection. (CO2, CO5)

5. (a) Explain the concept of abrasive perforation. (CO2, CO3, CO4)

(b) Differentiate between wellbore performance and choke performance.

(CO2, CO3, CO4)

(c) Explain the artificial lift method which can be used for volumes range from a low of 150 B/D. (CO2, CO3, CO4)

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B. TECH. (PE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

ALTERNATE ENERGY RESOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are various sources of "Renewable Energy". (CO1, CO6)
(b) Describe the various non-conventional sources relevant to India with focus on latest trends and global energy scenario. (CO1, CO6)
(c) Enumerate the criteria based on which energy can be classified. Give their availability, relative merits, demerits and their classification.
(CO1, CO6)
2. (a) Explain briefly the factors which affect the performance of a flat plate collector. (CO2)

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- (b) Calculate the values of R_b , R_r and R_d for an inclined surface facing due south, tilted at 30° with horizontal at 10 am for a location with latitude $20^\circ 51'$ on January 1. Take $\rho = 0.2$. (CO2)
- (c) Describe the principle of solar photovoltaic energy conversion with neat diagram. (CO2)
3. (a) Explain the various types of turbines considered for use in micro hydro resources and various types of controls available in turbines. (CO3)
- (b) Explain the various types of speed governors, generating systems and generators considered for use in micro hydro resources. (CO3)
- (c) Classify Hydropower Systems. What are the major advantages of mini/micro hydro resources? (CO3)
4. (a) Enumerate wind recording methods accepted by World Meteorological Organization. (CO4)
- (b) A propeller type horizontal shaft wind turbine has following wind characteristics : Speed of wind 10 m/s at 1 atm and 15°C . Turbine has a diameter of 120 m and its operating speed is 40 rpm at maximum efficiency.
- Calculate : (CO4)
- (i) Power density in the wind stream,
 - (ii) Maximum obtainable power density assuming $\eta = 40\%$,
 - (iii) Total Power produced in kW and
 - (iv) Torque and Axial thrust.
- (c) What are the basic components of wind energy conversion systems (WECS)? Explain them with a neat block diagram. (CO4)

(3)

5. (a) Following data is given for a family biogas digester suitable for output of five cows: the retention time is 20 days, temperature 30°C , dry matter consumed per day = 2 kg, biogas yield is 0.24 m^3 per kg. The efficiency of burner is 60%, methane proportion is 0.8. Heat of combustion of methane = 28 MJ/m^3 . Calculate : (i) the volume of biogas digester, (ii) the power available from the digester. (CO5)
- (b) What are the fuel properties of biogas ? Explain briefly the methods used to obtain energy from biomass. (CO5)
- (c) Explain with a schematic diagram the "Anaerobic digestion system".

(CO5)